Report Title: The Influence of Social Media Use on Loneliness: A Mediation Analysis of Rumination

Abstract

Although social media is often assumed to foster social connection, emerging evidence suggests it may also contribute to negative mental health outcomes, such as loneliness. Using a cross-sectional online survey, this study examined whether rumination mediates the relationship between social media use and loneliness in adults aged 18-50 who use social media daily. Measures included the Social Media Use Integration Scale (SMUIS), the Ruminative Response Scale-Short Form (RRS-SF), and the UCLA Loneliness Scale. A simple mediation model tested whether social media use predicted loneliness indirectly through rumination. Although social media use did not directly predict loneliness, it significantly predicted rumination, which in turn predicted higher loneliness. The indirect effect was significant, indicating a full mediation. These findings suggest that rumination (not social media use itself) may be the key psychological mechanism explaining increased loneliness. The results highlight the potential value of interventions that help individuals manage maladaptive thought patterns to support wellbeing.

Introduction

Although social media platforms are designed to connect people, it can paradoxically increase feelings of loneliness, particularly when use involves passive engagement or frequent exposure to idealised portrayals of others' lives. Social comparison theory (Festinger, 1954) helps explain why this happens: people naturally measure themselves against others, and when they compare themselves to those who seem to be doing better (known as upward social comparison) it can leave them feeling inadequate, lowering self-

esteem and negatively affecting emotional well-being. Empirical evidence supports this link; for example, Dibb and Foster (2021) found that Facebook users who engage more frequently in upward social comparison reported higher loneliness, partly due to rumination – getting stuck in a loop of negative thoughts about oneself. The study's 250 Facebook users self-reported, offering insight but limiting generalisability. Similarly, Xu and Li (2024) found that upward social comparison predicted higher social anxiety among college students, with rumination mediating this relationship. Although their use of a convenience undergraduate sample and a single-timepoint survey limit generalisability and casual conclusions, the study shows how persistent negative thinking can heighten the emotional impact of online comparisons.

Rumination plays a central role in how comparing ourselves to others can lead to negative emotions, as it involves getting caught up in repetitive, self-critical thoughts. Defined as repetitive, passive focus on negative feelings and experiences (Nolen-Hoeksema, 1991), rumination is linked to depression, anxiety, and loneliness. On social media, people may fixate on their perceived shortcomings compared with others, which can heighten negative emotions. Seabrook et al., (2016) and Feinstein et al. (2013) further found that rumination mediated the relationship between heavy social media use and poorer well-being, suggesting that rumination may be the mechanisms through which comparing ourselves online leads to increased loneliness. This highlights that addressing rumination could help reduce the harmful emotional impact of using social media.

Despite this evidence, gaps remain. Most research has focused on general social media use without examining the specific cognitive processes, such a rumination, that may explain why some users experience greater loneliness. The present study builds on previous research by directly examining rumination as a mediator, offering insight into the cognitive mechanisms through which social media use may impact emotional well-being in adults. Clarifying this

pathway can advance theoretical understanding and inform interventions, particularly for young adults, who are heavy social media users and especially susceptible to ruminating on social comparisons. Individual differences, such as mindfulness and self-esteem, may also influence these processes (Gu et al., 2022), but their role has not yet been fully integrated into models predicting loneliness.

The present study aims to test whether rumination mediates the relationship between social media use and loneliness in adults. We will examine a mediation model in which greater social media use predicts higher levels of rumination, which, in turn, contributes to increased feelings of loneliness. Based on this model, we hypothesise that social media use will be positively associated with rumination (H1), rumination will be positively associated with loneliness (H2), and social media use will have an indirect effect on loneliness through rumination (H3).

By testing this model, the study seeks to improve our understanding of how social media influences emotional well-being and the cognitive processes that link social media use to loneliness. Understanding rumination as a key link between social media use and loneliness could help design interventions that reduce unhelpful thinking patterns, supporting young adults to use social media in ways that benefit, rather than harm, their mental well-being.

Method

Design

This study employed a cross-sectional, online psychometric survey design. A simple mediation model was used to examine whether rumination mediated the effect of social media use (predictor) on loneliness (outcome). All data were collected anonymously, and participants were informed they could withdraw at any time.

Participants

Participants were recruited via convenience sampling through university networks, social media platforms and personal contacts. Inclusion criteria required participants to be aged 18-50 and to use social media for at least 30 minutes per day. The broad age range was chosen to capture variability in social media use across adulthood, though it may reduce representatives of specific age subgroups. Data exclusion criteria included incomplete responses, duplicate entries or non-serious report patterns. The final sample size reflected the pre-registered target of approximately 50 participants.

Materials

Social Media Use Integration Scale (SMUIS; Jenkins-Guarnieri et al., 2013).

This scale measures the extent to which social media is integrated into an individual's social routines and identity. It was chosen as it directly measures social media engagement, central to the hypothesis linking use to rumination and loneliness. Items are rated on a Likert scale, with higher scores indicating greater integration of social media into daily life. Internal reliability typically exceeds $\alpha = .80$.

Ruminative Response Scale – Short Form (RRS-SF; Treynor et al., 2003).

This 10-item scale assesses trait rumination, with items reflecting brooding and reflective pondering. It was selected as the mediator because it measures repetitive negative thinking, hypothesised to link social media use and loneliness. The RRS-SF is considered the gold-standard measure of rumination and has demonstrated strong reliability ($\alpha = .85$).

UCLA Loneliness Scale (Russel, 1996).

This widely used and well-validated scale assesses subjective loneliness and perceived social isolation. It was chosen to measure loneliness, the study's key outcome. Participants rate items such as "How often do you feel left out?". Reliability is consistently high ($\alpha \geq .80$).

Procedure

The study was hosted on Gorilla. Following informed consent, participants completed demographic questionnaire and the three scales in a fixed order, to ensure consistency across participants. Responses were recorded securely, and participants were debriefed at the end of the survey. Data were screened for missing values, duplicates, and response patterns to ensure quality; normality of scores was also checked before analysis.

Results

Descriptive Statistics

Descriptive statistics for all study variables are shown in Table 1. Social media use scores were moderately high (M = 3.32, SD = 0.90). Rumination scores were somewhat lower (M = 2.45, SD = 0.80). Loneliness scores were relatively low in the sample (M = 1.96, SD = 0.95).

Table 1

Descriptive Statistics and Correlations for Social Media Use, Rumination, and Loneliness

Variable	M	SD
1. Social media use	3.32	0.90
2. Rumination	2.45	0.80
3. Loneliness	1.96	0.95

Note. N = 50

Correlations

Correlational analyses indicated that social media use was positively correlated with rumination and loneliness. Rumination was also positively correlated with loneliness (all $p < .01$).

Mediation Analysis

A mediation analysis was conducted to assess whether rumination mediated the relationship between social media use and loneliness. The predictor (social media use) significantly predicted the mediator (rumination), a = 0.63, SE = 0.09, z = 7.44, p < .001. The mediator (rumination) significantly predicted the outcome (loneliness), b = 0.77, SE = 0.22, z = 3.56, p < .001. The direct effect of social media use on loneliness, controlling for rumination, was non-significant, c' = 0.04, SE = 0.16, z = 0.28, p = .778. The indirect effect via rumination was significant, ab = 0.49, SE = 0.16, z = 3.11, p = .002, 95% CI [0.23, 0.89]. The total effect of social media use on loneliness before including the mediator was significant, c = 0.53, SE = 0.12, z = 4.32, p < .001. These findings indicate full mediation, suggesting that the association between social media use and loneliness is accounted for by rumination.

As shown in Figure 1, rumination fully mediated the relationship between social media use and loneliness.

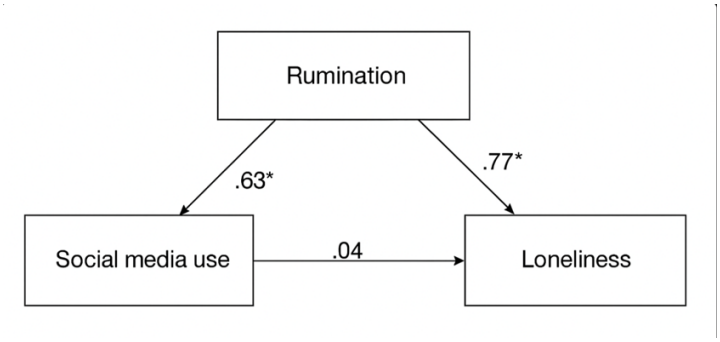

Figure 1

Mediation model illustrating the indirect effect of social media use on loneliness through rumination. Path coefficients are standardised. P < .05

Discussion

The present study examined whether rumination mediates the relationship between social media use and loneliness in adults aged 18-50. The findings partially supported the hypothesis: social media use significantly predicted rumination, and rumination significantly predicted loneliness. The direct pathway between social media use and loneliness was non-significant, while the indirect pathway was significant, indicating full mediation. This suggests that the link between social media use and loneliness is driven primarily by its impact on maladaptive processing, rather than by usage frequency alone. These results suggest that emotional responses to online environment are driven by how people interpret and engage with content than by their overall level of usage (Verduyn et al., 2017).

The findings align with literature highlighting the role of cognitive mechanisms in shaping social media's psychological effects. Dibb and Foster (2021) demonstrated that upward social comparison on Facebook predicted increased rumination, which subsequently heighted loneliness. Similarly, Xu and Li (2024) found that rumination mediated the association between social comparison and social anxiety, indicating that repetitive negative thinking is a

general mechanism linking social media experiences to poorer wellbeing. The current study extends this work by demonstrating that rumination mediated the association not only in the contexts of explicit social comparison cues but also in broader, everyday social media engagement. This supports Response Styles Theory (Nolen-Hoeksema 1991), suggesting that people who tend to ruminate are more prone to interpretating online content negatively, and that digital platforms may intensify this pattern by offering constant opportunities for comparison.

Broader research supports this cognitive interpretation. Verduyn et al. (2017) argued that passive browsing fosters self-focused, evaluative thought patterns resembling rumination, while Kross et al. (2013) reported that increased Facebook use predicted declines in affective wellbeing's overtime. Feinstein et al. (2013) similarly found that social media use predicted poorer psychological outcomes through increased rumination. However, their correlational design and reliance on student participants limit casual interpretation and reduce generalisability, reflecting broader issues in the social media and wellbeing literature. Nonetheless, the evidence suggests that focusing on maladaptive cognitive responses - via reduced social comparison, reappraisal, or mindfulness - may be more effective than reducing online use alone.

Despite these strengths, limitations remain. Because the design is cross-sectional, casualty cannot be inferred; loneliness, for example, may increase rumination and subsequently affect social media use. Longitudinal or experimental research is needed to determine temporal direction. Additionally, reliance on self-report measures risks the potential for response bias, particularly as ruminative individuals may recall experiences more negatively. The snowball sampling method may also limit representativeness by disproportionately recruiting participants with similar demographic or online behaviour profiles. Furthermore, the model

only included one mediator. Although rumination is a well-supported mechanism, factors like self-esteem, FoMO, emotional regulation, and perceived social support also shape wellbeing in digital settings and may contribute to a more complex cognitive network (Primack et al., 2017)

Future research should therefore consider multi-mediator or moderated mediation models. Age is a particularly promising moderator. Although the present sample included adults aged 18-50, age differences were not examined. Evidence suggests that younger adults may be more vulnerable to upward social comparison and rumination (Gu et al., 2022), whereas older adults may use social media more practically and experience fewer negative cognitive effects. Examining whether age moderates the link between social media and rumination, or between rumination and loneliness, could offer a more developmentally sensitive understanding of how social media affects wellbeing.

Overall, the present findings provide further evidence that rumination plays a central role in explaining why social media use is associated with loneliness. Emotional outcomes appear driven not by social media use itself, but how online content is processed – especially through repetitive negative thinking. These findings highlight the potential of interventions targeting maladaptive cognitive styles to reduce social media's psychological risks.

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Appendix

This project focused on the Wolfpack Project's question: How can social media, which is designed to connect people, still make young adults feel lonely? Using psychological research and our own data, we explored whether rumination (repetitive negative thinking) helps explain why heavier social media use is linked to greater loneliness.

Across studies, young adults who spend more time online tend to compare themselves to others, overthink social interactions, and dwell on negative feelings. This pattern aligns with wider evidence showing that social media can trigger cycles of self-criticism and worry, which increase loneliness rather than reduce it. Our project tested this using three scales: social media integration (how central social media is in daily life), rumination, and loneliness.

Our results showed that higher social media use was associated with greater rumination. And in turn, higher rumination was linked to higher loneliness. This suggests that it may not be social media alone that leads to loneliness, but what happens in the mind while using it. Young adults who dwell on conversations or compare themselves to online peers may become increasingly isolated. This means that rumination is a promising psychological mechanism for understanding loneliness in the young people you support.

These findings are useful for the Wolfpack Project because they point to specific areas where intervention may help. Rather than focusing only on reducing screen time, support could target how young adults think about and interpret their online experiences. Teaching them that algorithms amplify idealised content can reduce overthinking and self-criticism. Teaching simple strategies such as limiting doomscrolling, unfollowing accounts triggering comparison, and scheduling offline activities may also reduce ruminative thinking.

However, the results should be interpreted with caution. Our study used self-report questionnaires, which rely on our participants being honest and aware of their own habits.

The design was correlational, meaning we cannot say for certain that social media causes rumination or loneliness- only that they're related. Young adults who already feel lonely may also turn to social media in unhelpful ways. Additionally, our sample consisted of university students, so experiences may differ for other young adults across Nottinghamshire, especially those facing additional stressors such as housing insecurity, unemployment or mental health conditions.

Even with these limitations, the findings offer clear and realistic implication for your work. When supporting young adults, consider asking not just how much they use social media but how they feel during and after using it. Encouraging reflective conversations about comparison, online pressure, and self-criticism may help reduce loneliness. Workshops could include activities that teach emotional regulation, setting healthy boundaries online, and recognising when rumination is starting. Peer-support groups may also provide safe spaces to discuss online experiences only.

Finally, your organisation could signpost individuals to services such as Mind or YoungMinds, which offer accessible information and tools for managing mental health challenges linked to social media use. With these evidence based approaches, the Wolfpack Project can continue empowering young people to develop resilience and more positive online habits.